

AARHUS UNIVERSITY

DEPARTMENT OF NERDS (MATH)

INTRODUCTION TO ALGEBRA 1

If this is the title, please don't read
me...

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Goofy ah Math is scawy oWo

Exercise: In the following, n denotes a natural number.

1. Show, by induction on n , that

$$\sum_{k=1}^n \frac{1}{k^2} \leq 2 - \frac{1}{n}.$$

The base case for $n = 1$

$$\text{LHS: } \frac{1}{1^2} = 1 \quad \text{RHS: } 2 - \frac{1}{1} = 1$$

the base case is satisfied since $1 \leq 1$

Next step is the hypothesis for induction ThingyMaBob, which is to assume it holds for some positive number $\Omega = n$ (Queue introduction to the Omegaverse) #FanficForLife
The Goofy ah Ω goes to town #Purr

$$\sum_{k=1}^{\Omega} \frac{1}{k^2} \leq 2 - \frac{1}{\Omega}$$

Now it is time for marriage, which is to prove it for $\Omega + 1$, the next number in the chain, so the actual induction step.

But remember as a true Alpha, we know the left side is just the summation:

$$\frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{\Omega^2}$$

This means the next one in the series is just $\frac{1}{(\Omega+1)^2}$ so the summation for $\Omega + 1$ becomes

$$\sum_{k=1}^{\Omega+1} \frac{1}{k^2}$$

And for the right hand side, the next number is just added as so #HowlingForTheMoon

$$\sum_{k=1}^{\Omega+1} \frac{1}{k^2} = 2 - \frac{1}{\Omega} + \frac{1}{(\Omega+1)^2}$$

From here it's a matter of transformation (under the fullmoon #WerewolfBaddie)

$$2 - \frac{1}{\Omega} + \frac{1}{(\Omega + 1)^2} \quad (1)$$

$$2 - \frac{1 \cdot (\Omega + 1)^2 + \Omega \cdot 1}{\Omega(\Omega + 1)^2} \quad (2)$$

$$2 - \frac{(\Omega + 1)^2 + \Omega}{\Omega(\Omega + 1)^2} \quad (3)$$

$$2 - \frac{\Omega^2 + 2\Omega + 1 + \Omega}{\Omega(\Omega + 1)^2} \quad (4)$$

$$2 - \frac{\Omega^2 + \Omega}{\Omega(\Omega + 1)^2} - \frac{2\Omega + 1}{\Omega(\Omega + 1)^2} \quad (5)$$

$$2 - \frac{\Omega(\Omega + 1)}{\Omega(\Omega + 1)^2} - \frac{2\Omega + 1}{\Omega(\Omega + 1)^2} \quad (6)$$

$$2 - \frac{1}{\Omega + 1} - \frac{2\Omega + 1}{\Omega(\Omega + 1)^2} \quad (7)$$

2. Conclude that for all natural numbers n , it holds that

$$\sum_{k=1}^n \frac{1}{k^2} < 2.$$

3. Consider whether the conclusion in 2 can be shown directly by induction on n .